

Securing the Quantum Channel

A Simulated BB84 Quantum Key Distribution Pipeline:
Key Exchange, Eavesdropper Detection, and Secure Key Distillation

Qcrypt — Team 10

Track 3 — Technical Report

Abstract

We implement and analyze an end-to-end BB84 quantum key distribution pipeline: circuit-level state preparation and measurement, public sifting, an intercept-resend eavesdropper model, three independent channel-noise sources, adaptive Cascade error reconciliation, and Toeplitz-hash privacy amplification. We quantify the quantum bit error rate (QBER) as a function of both eavesdropping strength and channel noise, justify an 11% operational abort threshold against the 25% theoretical intercept-resend bound, and report the resulting final-key-rate-vs-QBER curve as the central benchmark of the full pipeline, compared against the asymptotic bound $1 - 2H_2(\text{QBER})$.

1 Protocol Description

We implement BB84 prepare-and-measure quantum key distribution end-to-end in Qiskit, simulated on the Aer statevector backend, together with a full classical post-processing pipeline (sifting, Cascade-based error reconciliation, and Toeplitz-hash privacy amplification). The system models an active intercept-resend eavesdropper (Eve) and three independent channel-noise sources (depolarizing bit flips, photon loss, detector dark counts), so that eavesdropping and ordinary noise can be told apart.

1.1 Key exchange

- Alice generates N random bits and N random encoding bases (Z or X) and prepares each qubit accordingly: bit 1 $\rightarrow X$ gate; X -basis $\rightarrow$ additional H gate.
- The qubit is sent over a channel (ideal, noisy, or Eve-intercepted, depending on the experiment).
- Bob independently chooses a random measurement basis for each qubit (H before measurement if X -basis) and measures in the computational basis.
- **Sifting:** Alice and Bob publicly compare (not reveal) their basis choices and keep only the bits where their bases matched, discarding the rest.

Over a single 30-qubit run the sifted key retained 14 of 30 bits (46.7%); averaged over repeated trials at increasing raw-key length, the sifting efficiency converges tightly to the theoretical 50% expected for two independently random 2-basis choices (Fig. 1).

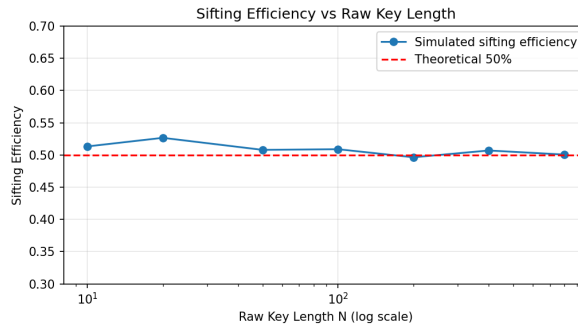


Figure 1: Sifting efficiency converges to the theoretical 50% as raw key length grows; small- N runs show larger statistical variance.

Raw key length N	10	20	50	100	200	400	800
Avg. sifting efficiency	51.3%	52.7%	50.8%	50.9%	49.7%	50.7%	50.1%

Table 1: Sifting efficiency vs. raw key length, averaged over 30 trials per point.

2 Eavesdrop Detection

Eve is modeled as an intercept-resend attacker: with probability p (the attack strength) she intercepts a given qubit, measures it in a randomly chosen basis, and resends a freshly prepared qubit consistent with her measurement outcome to Bob. Because she measures in the wrong basis roughly half the time, this attack introduces detectable errors into the sifted key even when Alice’s and Bob’s own equipment is otherwise perfect.

For a qubit Eve intercepts, she guesses the wrong basis with probability $1/2$; when she does, and Bob later happens to measure in Alice’s original basis, Bob’s result is uncorrelated with Alice’s bit and disagrees half the time. This yields an expected error rate of $0.5 \times 0.5 = 25\%$ on intercepted-and-basis-matched qubits — the well-known BB84 intercept-resend bound. Since Eve attacks a fraction p of all qubits, the expected sifted-key QBER is $\text{QBER} \approx 0.25p$.

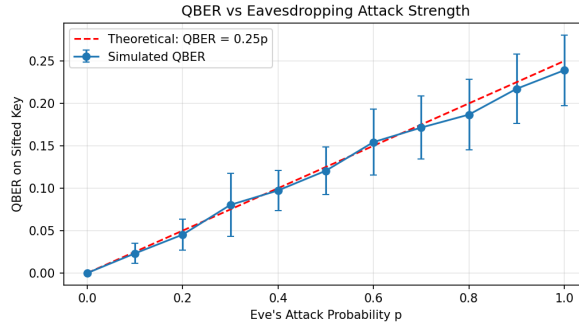


Figure 2: Simulated QBER vs. Eve’s interception probability p , averaged over repeated trials (error bars = 1σ across trials), against the theoretical $\text{QBER} = 0.25p$ line.

The simulation tracks the theoretical line closely across the full sweep (e.g. at $p = 0.5$, simulated $\text{QBER} = 12.1\%$ vs. theoretical 12.5%), confirming the intercept-resend model is implemented correctly.

2.1 Abort threshold

We adopt $\text{QBER} = 11\%$ as the abort threshold, consistent with commonly cited practical BB84 thresholds in the literature once reconciliation and privacy-amplification costs are accounted for (below the raw 25% single-qubit intercept-resend bound, which is only reached in the limit $p \rightarrow 1$).

Attack strength p	0.1	0.3	0.5	0.7	0.9
Simulated QBER	2.3%	8.0%	12.1%	17.1%	21.7%
Exceeds 11% threshold?	No	No	Yes	Yes	Yes
Exceeds 25% bound?	No	No	No	No	No

Table 2: Sifted-key QBER vs. attack strength, checked against the operational and theoretical bounds.

3 Noise Model & Channel Analysis

A real deployment must distinguish genuine channel noise from an active adversary. We separately model: (a) *depolarizing noise* — random bit flips at rate q applied to Bob’s measured bits; (b) *photon loss* — each transmitted qubit is lost (and simply discarded, not corrupted) with probability equal to the loss rate; and (c) *detector dark counts* — spurious clicks that overwrite Bob’s outcome with a uniformly random bit at rate equal to the dark-count rate.

Depolarizing noise and dark counts both raise the sifted-key QBER approximately linearly with their respective rates, and both can cross the 11% abort threshold at noise levels well within realistic hardware specifications (roughly 8–10% depolarizing probability or ~ 18 –20% dark-count rate in this simulation). This is the central practical risk in QKD deployment: a sufficiently noisy but Eve-free channel can trigger a false eavesdropping alarm. Photon loss behaves qualitatively differently — it shrinks the sifted key length without introducing any bit errors, so it does not by itself raise the QBER (Table 3).

This clean separation — loss affects key rate, not error rate; noise and eavesdropping both affect QBER similarly — is exactly why an operational system must complement the QBER-based abort rule with independent channel

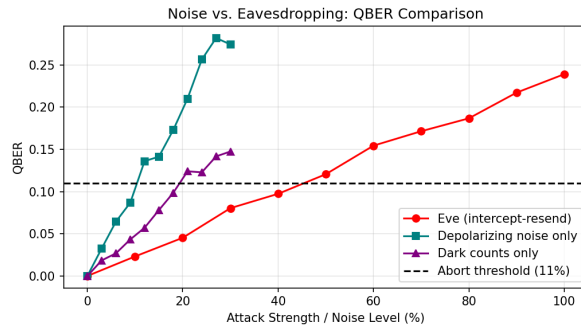


Figure 3: QBER as a function of attack/noise strength for Eve’s intercept-resend attack vs. two pure-noise sources (depolarizing noise, detector dark counts), with the 11% abort threshold marked.

Photon loss rate	0%	12%	24%	36%	48%	60%
Sifted bits retained (of ~163 baseline)	163	148	133	99	77	49
QBER	0%	0%	0%	0%	0%	0%

Table 3: Photon loss reduces sifted key length but leaves the QBER unchanged.

characterization (e.g. calibrating expected dark-count and loss rates from known-Eve-free maintenance windows) to avoid false-positive aborts on a merely noisy but secure link.

4 Secret Key Distillation

4.1 Error reconciliation (Cascade)

Discrepancies remaining after sifting are corrected using a simplified adaptive Cascade protocol: the sifted key is randomly permuted, split into blocks, and parity bits are compared block-by-block; any block with mismatched parity is binary-searched down to the single differing bit, which is corrected in place. The block size is grown across passes and the process repeats until Alice’s and Bob’s keys agree bit-for-bit (or a pass limit is reached). Each publicly disclosed parity bit is counted as one bit of information leaked to Eve.

As expected for a simplified (non-optimal) Cascade implementation, the observed leakage tracks above the Shannon-entropy lower bound, with the gap widening at higher QBER — a realistic reconciliation-efficiency penalty relative to an idealized error-correcting code operating at the Shannon limit. Trials whose observed QBER exceeds the 11% abort threshold are discarded before reconciliation is attempted.

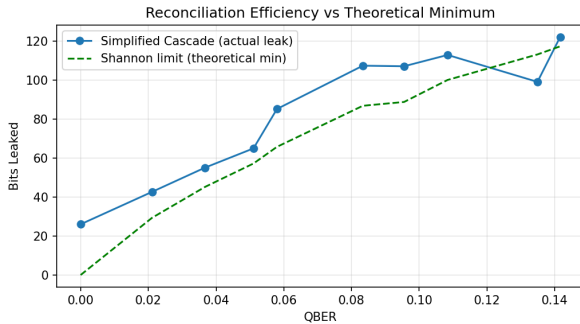


Figure 4: Bits leaked during Cascade reconciliation vs. the Shannon-entropy lower bound $n \cdot H_2(\text{QBER})$.

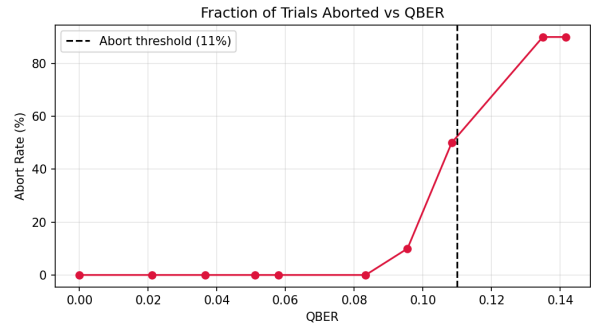


Figure 5: Fraction of trials aborted at the 11% threshold, vs. observed QBER.

4.2 Privacy amplification

After reconciliation, a Toeplitz-matrix universal hash compresses the reconciled key down to a shorter, provably secure final key. The output length is chosen using a security-margin calculation grounded in the Leftover Hash Lemma (Renner, 2005; Tomamichel et al., 2011): the final length subtracts (i) the reconciliation leakage, (ii) an entropy-based bound on Eve’s information from the observed QBER ($n \cdot H_2(\text{QBER})$), and (iii) an explicit finite-size security margin

$$\text{margin} = 2\log_2(1/\varepsilon_{\text{PA}}) + \log_2(1/\varepsilon_{\text{EC}}) \quad (1)$$

parameterized by the desired privacy-amplification and error-correction failure probabilities ε .

4.3 End-to-end demonstration

With a 600-qubit raw transmission and Eve intercepting a fraction $p = 0.2$ of qubits: the sifted key was 316 bits (observed QBER 4.43%, below the 11% abort threshold), reconciliation leaked 118 bits, and after subtracting Eve’s entropy bound and a 59.8-bit security margin ($\varepsilon = 10^{-6}$), the final secure key was **55 bits** long. Alice’s and Bob’s final hashed keys matched bit-for-bit, confirming end-to-end correctness of the pipeline.

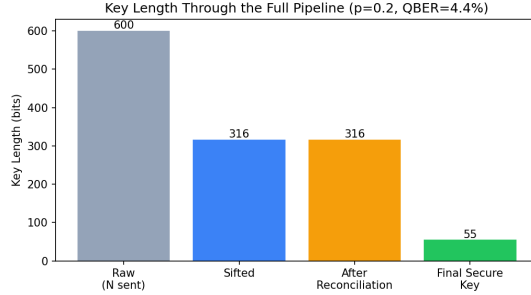


Figure 6: Key length shrinking through the pipeline: raw transmission → sifted key → post-reconciliation → final secure key (privacy-amplified).

4.4 Security-parameter trade-off

Tightening the target security parameter ε (i.e. demanding a stronger composable security guarantee) increases the required security margin and therefore shrinks the final key for a fixed input, making explicit a trade-off that a flat, arbitrary constant would hide:

ε ($= \varepsilon_{PA} = \varepsilon_{EC}$)	10^{-3}	10^{-6}	10^{-9}	10^{-12}
Security margin (bits)	29.9	59.8	89.7	119.6
Final key length (bits)	85	55	25	0

Table 4: Final key length shrinks as the target security parameter ε is tightened, for the fixed 316-bit reconciled key of Section 4.3.

5 Final Key Rate — Central Pipeline Result

The final key rate (secure output bits per raw qubit sent) as a function of the observed QBER is the single summary statistic for the whole pipeline’s performance, combining sifting, reconciliation cost, and privacy-amplification shrinkage.

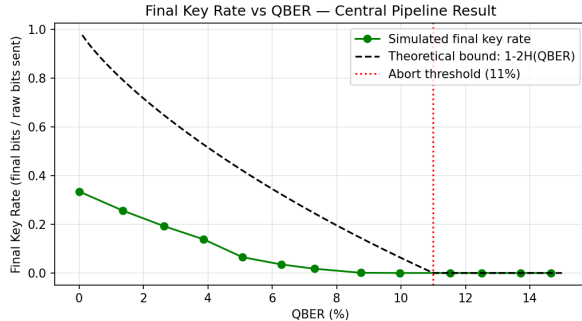


Figure 7: Final key rate vs. QBER, vs. the asymptotic bound $1 - 2H_2(\text{QBER})$. Rate hits zero at the 11% abort threshold.

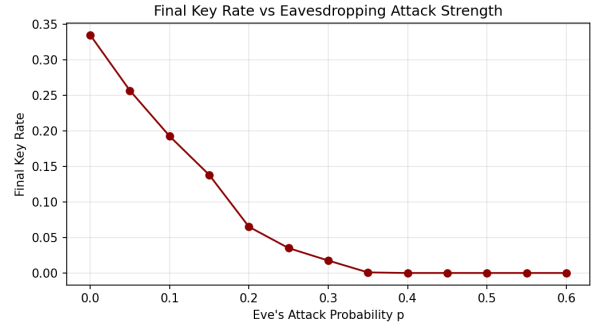


Figure 8: Final key rate vs. Eve’s attack probability p — the attacker-facing view of the same result.

The simulated key rate sits below the asymptotic theoretical curve at every QBER, which is expected: the theoretical curve $1 - 2H_2(\text{QBER})$ assumes an idealized, Shannon-efficient reconciliation code and no finite-size security margin, whereas our pipeline pays a real (larger) reconciliation cost (Fig. 4) and an explicit finite-key security margin (Section 4.4). Both curves reach zero key rate in the same QBER region, and our abort rule conservatively cuts off key generation at 11% QBER — well before the asymptotic bound would reach zero — trading some throughput for a wider security margin against finite-key statistical fluctuations.

6 Benchmarking & Security Discussion

6.1 Comparison against theoretical bounds

- **Sifting efficiency:** simulated $\approx 50\%$ for $N \gtrsim 100$, matching theory exactly.
- **Intercept-resend QBER:** simulated QBER tracks $0.25p$ closely across the full $p \in [0, 1]$ sweep.
- **Reconciliation cost:** simulated leakage exceeds the Shannon bound $n \cdot H_2(\text{QBER})$, as expected for a simplified, non-adaptive-optimal Cascade implementation.
- **Key rate:** simulated rate falls below and reaches zero earlier than the asymptotic bound $1 - 2H_2(\text{QBER})$, consistent with real reconciliation inefficiency plus a finite-key security margin.

6.2 Finite-key effects

Sampling QBER from small sifted-key sizes shows substantially higher variance than large ones, at a fixed attack strength ($p = 0.3$):

Sifted sample size	50	200	1000
Mean QBER	8.85%	7.60%	7.34%
Std. dev. of QBER	5.83%	2.94%	1.12%

Table 5: QBER estimation variance shrinks with sample size at fixed attack strength $p = 0.3$.

At $N = 50$ the standard deviation is roughly $5\times$ that at $N = 1000$. In a real deployment this means an abort decision made on a short block risks both false negatives (a real attack producing a QBER sample that dips under threshold by chance) and false positives (a benign, noisy channel producing a sample that spikes over threshold). This is the core motivation for finite-key security proofs, which replace the asymptotic bound $1 - 2H_2(\text{QBER})$ with confidence-interval-adjusted bounds that explicitly account for sample size — our pipeline’s ε -parameterized security margin (Section 4.4) is a simplified stand-in for this effect, not a full finite-key proof.

6.3 What would break this implementation

- **Finite-key statistics:** our privacy-amplification margin is derived from the Leftover Hash Lemma but does not perform full finite-key QBER estimation (e.g. Chernoff-bound confidence intervals on the sampled error rate), so short keys are more vulnerable to both false aborts and under-conservative key rates than a rigorous finite-key security proof would allow.
- **Imperfect single-photon sources:** this simulation assumes ideal single-photon-per-pulse preparation. Real implementations typically use weak coherent pulses, which occasionally emit more than one photon per pulse and are vulnerable to photon-number-splitting (PNS) attacks, where Eve siphons off extra photons without disturbing the transmitted bit and so evades QBER-based detection entirely. Countermeasures (decoy-state protocols) are not modeled here.
- **Detector side-channel attacks:** our model treats Bob’s measurement as an ideal, basis-blind black box. Real detectors can leak basis information through timing, efficiency mismatch, or blinding/bright-light attacks (e.g. detector-control attacks), which can let Eve learn Bob’s basis or force deterministic outcomes without raising the QBER — a known real-world BB84 vulnerability class outside the scope of this simulation.
- **Basis-independent loss assumption:** our photon-loss model is basis- and bit-independent; a real channel with basis-dependent loss could be exploited to bias which qubits survive to reach Bob.

7 Deliverables Summary

- **Code:** Qiskit/Aer circuit-level BB84 state preparation and measurement, classical sifting, intercept-resend Eve model, three independent noise models, adaptive Cascade reconciliation, Toeplitz-hash privacy amplification, and full key-rate sweep harness (see accompanying notebook).
- **This technical report:** protocol description, QBER-vs-attack-strength analysis (Fig. 2), noise-vs-eavesdropping separation (Fig. 3), reconciliation/privacy-amplification pipeline (Figs. 4–6), key-rate-vs-QBER central result (Figs. 7–8), and security discussion (Section 6).
- **Results dashboard values:** sifting efficiency ($\sim 50\%$), QBER curves with/without Eve, final key rate curve, and abort-threshold justification (11%, vs. 25% theoretical single-qubit bound) — all reproduced in the tables and figures above.